RP-GHOST

An Oracle-Seeded Cybernetic Computational Architecture & Theoretical Evaluation

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Domain: Cybernetics · Optoelectronics · Information Theory

Type: Full Technical Evaluation & System Specification

ABSTRACT

RP-GHOST is an experimental cybernetic architecture combining camera-based environmental sensing, optoelectronic logic, photon-electron statistical modeling, temporal coordination, and independent computational evaluation. The physical device uses an optical acquisition system (USB camera or IR laser-reflector bench) to observe environmental and quantum fluctuations. Changes are processed through parallel probabilistic (Driver A) and deterministic (Driver B) decision pathways. Their outputs are combined via charge-integrating threshold links, logic gates, and 74HC74 clocked state retention mechanisms. This paper presents a complete evaluation spanning computational proof-of-work search complexity, optoelectronic photon statistics, single-photon emission models, and physical feedback protocols.

1. COMPUTATIONAL COMPLEXITY & PROOF-OF-WORK ANALYSIS

The core distinction in evaluating the architecture's mining and search performance is **exponential search complexity vs. oracle processing time**. The oracle can generate its 64-bit input in linear time `O(n)`, but Bitcoin SHA-256d proof-of-work still requires finding a nonce whose hash falls below the network target. The codebase explicitly separates these two operations.

1.1 Difficulty and Search Time Scaling

The configuration specifies local testing difficulty via:

```
TEST_DIFFICULTY = 2 ** (TEST_BITS - 32)
```

This converts the requested number of leading-zero-equivalent bits into Bitcoin difficulty. The expected number of hashes required to satisfy a target defined by *k* bits is given by:

$$E[H] = 2^{TEST_BITS}$$

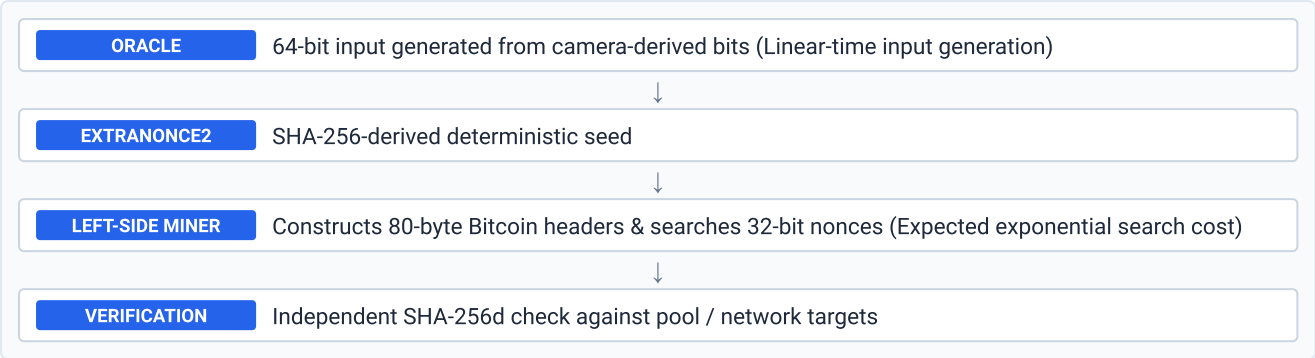
Assuming a processing baseline of 500,000 SHA-256d hashes per second, search times scale exponentially as follows:

Test Bits	Expected Hashes (E[H])	Expected Search Duration
20 bits	1,048,576	2.1 seconds
24 bits	16,777,216	33.6 seconds
28 bits	268,435,456	8.9 minutes
32 bits	4,294,967,296	2.4 hours
40 bits	1,099,511,627,776	25.5 days
47 bits	140,737,488,355,328	8.9 years
48 bits	281,474,976,710,656	17.8 years

Note: These represent mathematical expectation values, not guaranteed completion upper bounds.

2. WHY THE ORACLE CANNOT SOLVE THE COMPLEXITY DIFFERENCE

The system architecture separates seed generation from header search. The oracle determines the starting input, not the probability of satisfying the mathematical target:



Even if the oracle generates its input instantaneously (in `O(n)` linear time), a random SHA-256d hash output retains an identical probability of satisfying a given difficulty target.

3. LOCAL TEST MODE VS. PRODUCTION TARGET VERIFICATION

The implementation incorporates a local testing override mechanism:

```
TEST_MODE = True
TEST_BITS = 48

# Target selection logic
return max(
    difficulty_to_target(TEST_DIFFICULTY),
    cls.pool_share_target(job)
)
```

This logic compares local testing thresholds against actual pool thresholds and selects the easier target. Consequently:

- Local test shares can be found relatively quickly when difficulty parameters are reduced.
- Actual pool shares must strictly satisfy the pool's real target.
- Network blocks must satisfy the Bitcoin network target.
- Local test shares are not valid Bitcoin pool shares merely because the local verifier accepts them.

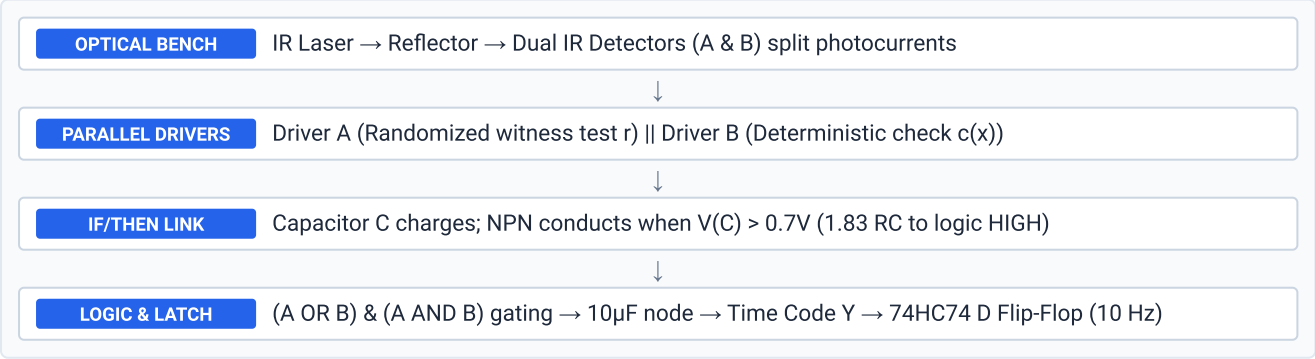
4. ARCHITECTURAL LIMITATION & MATHEMATICAL FORMULATION

The oracle functions as a physical seed generator rather than a proof-of-work accelerator. Linear-time oracle operation does not imply linear-time SHA-256d search complexity:

$$T_{total} = T_{oracle} + T_{search} \quad \text{where} \quad T_{oracle} = O(n) \quad \text{and} \quad E[T_{search}] \propto 2^{k}$$

5. OPTOELECTRONIC RP DECISION CIRCUIT ARCHITECTURE

The physical device models a hardware realization of a Randomized Polynomial-time (RP) decision procedure with one-sided error. An IR laser emits light across an optical bench to two detectors (A and B), driving parallel execution paths.



5.1 Threshold Dynamics & Transistor Kinetics

Driver B charges capacitor *C* through drive resistance *R*. The voltage follows:

$$V(C) = V_s (1 - e^{-t / RC})$$

For V_s = 5V, conduction begins at V(C) = 0.7V (t = 0.15 RC). However, standard 74HC logic input requires $V_{in} \approx 3.5V$ for logic HIGH, reaching valid input voltage at **1.83 RC**. The capacitor integrates charge across N = (C · 0.7V)/e electrons, reducing relative fluctuations to 1 / √N (≈ 0.05% for C = 1 pF), establishing deterministic switching.

6. PHOTON STATISTICS & ELECTRON REGULATING MECHANICS

Manipulating electron streams directly alters photon emission statistics. The correlation between electron Fano factor F_e and detected photon Fano factor F_{ph} under quantum efficiency η is governed by:

$$F_{ph} = 1 - \eta(1 - F_e)$$

Regime	Electron Statistics (F_e)	Photon Statistics (F_{ph})	Mandel Q / $g^{(2)}(0)$ Behavior
Random (Driver A)	$F_e = 1$ (Poissonian)	$F_{ph} = 1$ (Poissonian)	Q = 0, $g^{(2)}(0) = 1$ (Standard laser shot noise)
Regulated (Driver B)	$F_e = 0$ (Sub-Poissonian)	$F_{ph} = 1 - \eta$ (Sub-Poissonian)	Q < 0, $g^{(2)}(0) < 1$ (Antibunched single-photon emission)
Bunched (Avlanche/Traps)	$F_e > 1$ (Super-Poissonian)	$F_{ph} > 1$ (Super-Poissonian)	Q > 0, $g^{(2)}(0) > 1$ (Thermal / bunched photon arrivals)

7. PROTOCOL EXECUTION & CLOSED TIMELIKE CURVE (CTC) ANALYSIS

The complete physical trial operates as a 6-step optoelectronic protocol:

1. **Preparation:** Parameter x, laser drive, and time code Y set counting windows.
2. **Energy Switching:** Electrons drop $E_2 \rightarrow E_1$ emitting photons; detector absorption lifts electrons across bandgaps.
3. **Threshold Integration:** Driver A samples noisy signal; Driver B charges C until NPN conducts.
4. **Gate Evaluation:** AND gating enforces dual agreement (preserving one-sided error).
5. **Measurement Latch:** 10µF node filters glitches; 74HC74 latches output Q = SELECT on 10 Hz clock edge.
6. **Repetition:** k independent trials reduce false reject probability down to $(1/2)^k$.

7.1 Closed Timelike Curves (CTCs) vs. Electrical Feedback Loops

In the standard hardware implementation, signal flow is strictly **feed-forward** (no feedback path from Q to the drivers or pump). If a feedback wire were introduced from Q back to the input, it would form an ordinary time-delayed electrical loop.

In theoretical physics, Deutsch's model for quantum operations near Closed Timelike Curves (CTCs) enforces a consistency condition on the state ρ_{CTC}:

$$\rho_{CTC} = Tr_{other} [U (\rho_{other} \otimes \rho_{CTC}) U^{\dagger}]$$

While a CTC resolves grandfather-type paradoxes (e.g. x = NOT x) instantaneously via a 50/50 mixed state density matrix, a physical circuit with an inverted feedback loop (wiring Q-bar to D) resolves state contradiction through a **clock delay of T = 100 ms**, producing a stable 5 Hz output oscillation rather than a temporal fixed-point state.

8. SCIENTIFIC INTERPRETATION & CONCLUSION

RP-GHOST unifies physical-to-computational feedback, optoelectronic photon regulation, and formal randomized algorithm verification. The architecture establishes that while physical sensing and optical statistics can yield ultra-low noise deterministic gating and zero-false-accept logic, fundamental computational complexity classes (such as SHA-256d exponential proof-of-work search) remain strictly governed by mathematical difficulty boundaries.

APPENDIX A

Circuit Implementation & Circuit Logic Specification (camera build)

Source: rp_serial_circuit_camera.py **Scope:** complete signal chain, gate logic, timing, error analysis, and mapping to the paper (§5–§7)

ABSTRACT. This appendix specifies the decision circuit exactly as implemented in the reference code. Every camera frame is one trial of an RP-style (one-sided error) circuit: motion detection produces an instance x; Driver A runs a randomized Miller-Rabin witness test seeded by the motion pattern; Driver B runs a deterministic check through a capacitor/NPN link; the outputs are combined by tied OR/AND gates, filtered by a 10 µF node, gated by a time code Y, and latched by a 74HC74 on the frame edge. The final output SELECT is the OR of Q over the last k frames. The circuit never accepts an x outside the language L = {odd composite numbers}; time code Y sets the error exponent, not the sidedness.

A.1 Signal chain

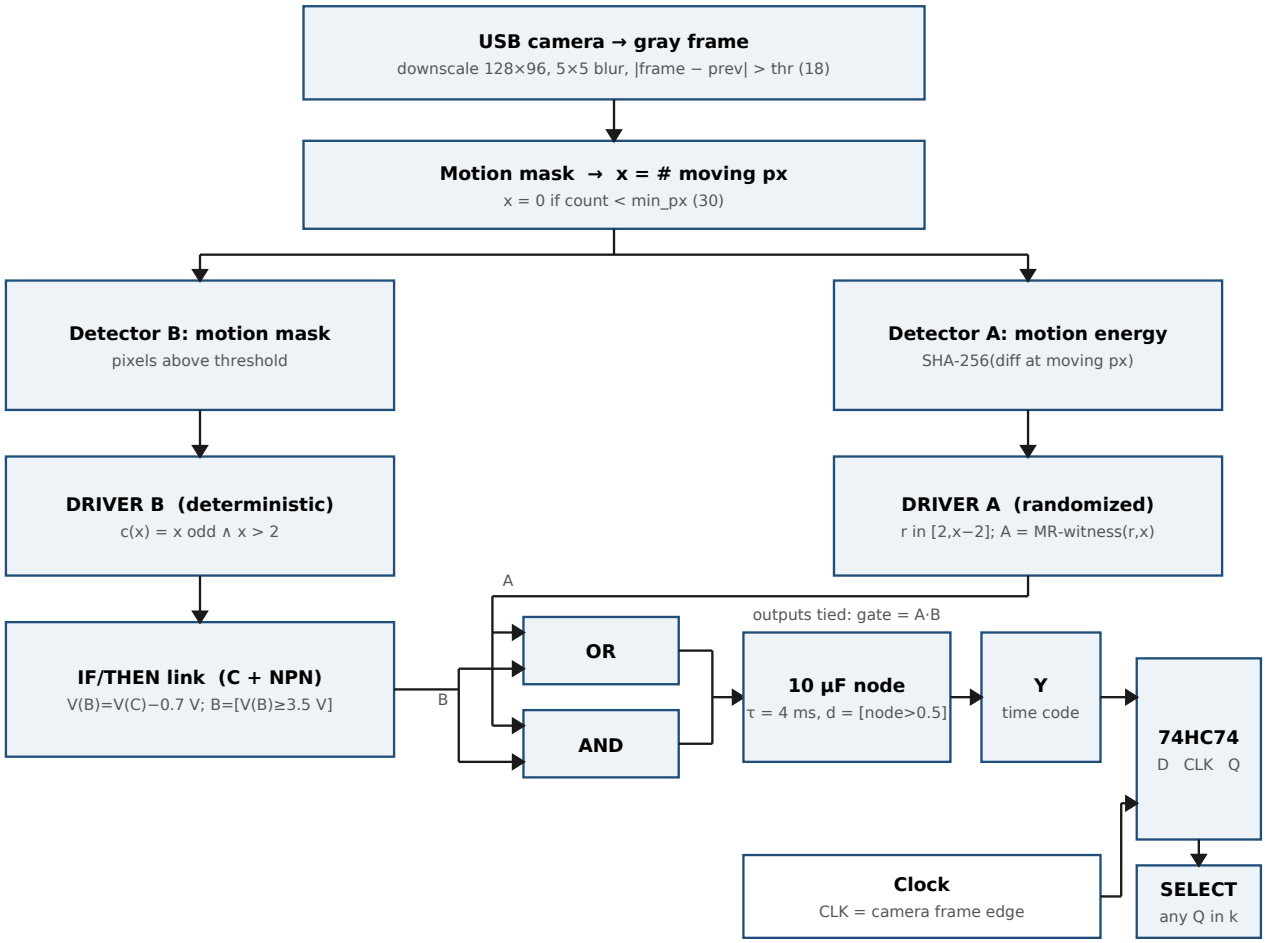


Figure A1. Block schematic from the code. Wires carry binary logic except the B link (analog V(C), V(B)) and the node (analog, thresholded at 0.5).

A.2 Constants and defaults

Symbol	Value	Meaning
SW × SH	128 × 96	Processing resolution after area downscale and 5×5 Gaussian blur
thr	18 (4–60)	Pixel-difference threshold for the motion mask
min_px	30 (5–800)	Minimum moving pixels for an instance; below it x = 0
WARMUP	8 frames	Frames ignored while auto-exposure settles
VS	5.0 V	Driver B supply / output voltage
V_BE	0.7 V	NPN turn-on (emitter-follower drop)
V_IH	3.5 V	74HC input-high threshold at 5 V
RC_B	4 ms	IF/THEN link time constant (R·C)
TAU_NODE	4 ms	10 µF node low-pass time constant
LOGIC_TIME	1.83 RC ≈ 7.3 ms	RC·ln(VS/(VS − V_IH − V_BE)) = RC·ln 6.25
k	5 (1–30)	Repetition window for SELECT
Y	"1111"	Time code; bin = Y[frame index mod len(Y)]
dt	clamped 1 ms–250 ms	Measured frame period; it is the integration time of the link and node

A.3 Stage-by-stage logic

Stage 0: sensing and instance

`diff = |small − prev|`; `mask = diff > thr`; `x = count(mask)` if count ≥ min_px, else 0. The witness source is `SHA-256(diff at moving pixels)`, so r is as unpredictable as the movement. No sensor noise is injected. The same digest is mixed into the factoriser's entropy pool.

Ground truth (used only for scoring, never by the circuit): $x \in L \Leftrightarrow$ motion $\wedge$ $x \geq 9$ $\wedge$ x odd $\wedge$ x has a divisor in $3..\lfloor\sqrt{x}\rfloor$.

Stage 1: Driver A (randomized witness test, Detector A = motion energy)

```
if motion and x > 4 and x odd:
    r = 2 + rand_int mod (x − 3)         # r in [2, x−2]
    A = MillerRabinWitness(r, x)
else: r = 0; A = 0
```

A = 1 only if r is a witness that x is composite. **For any prime, A = 0 for every r**; likewise for even x, $x \leq 4$ and no-motion frames. This is the one-sided-error property. For an odd composite, the witness density p is ≥ 3/4 (Rabin); the code computes the exact density per x and checks $p \geq 1/2$.

Stage 2: Driver B and the IF/THEN link (Detector B = motion mask)

c(x) = motion $\wedge$ $x > 2$ $\wedge$ x odd. This is a necessary condition for L: every member passes, but so does every odd prime, so B alone cannot decide L. When c(x) = 1, C charges from 0 through R for one frame period:

```
V(C)(t) = VS · (1 − exp(−t/RC)),   V(B) = V(C) − 0.7 V,   B = [ V(B) ≥ 3.5 V ]
threshold:  V(C) ≥ 4.2 V   ⇒   t = RC · ln(5/0.8) = 1.83 RC ≈ 7.3 ms
```

At 30 fps (dt ≈ 33 ms): t/RC ≈ 8.3, V(C) ≈ 4.998 V, V(B) ≈ 4.30 V, so B = 1. B can miss a member of L only if dt < 7.3 ms (above ≈ 137 fps); the analysis counts such events as *link_miss*. The 0.7 V conduction point is at t = 0.15 RC, as in paper §5.1.

Stage 3: gates

```
orv  = A | B
andv = A & B
gate = orv & andv        # outputs tied  →  (A+B)·(AB) = AB
```

Tying the OR and AND outputs reads as AND, enforcing dual agreement (randomized test says composite, deterministic test says odd and > 2). Because A = 1 already implies c(x) = 1, the AND with B is redundant except in the dt < 7.3 ms case.

Stage 4: 10 µF node, time code Y, 74HC74

```
node  = gate · (1 − exp(−dt/τ)),  τ = 4 ms ;   d_in = [ node > 0.5 ]
y_bin = Y[idx mod len(Y)] | added                 # 'added' = one earned bin, consumed once
Q     = d_in & y_bin          (gating)      Q = d_in | y_bin   (force mode, Thm 4 demo)
SELECT = OR( Q over the last k frames )
```

The node passes a steady 1 once dt > τ ln 2 ≈ 2.8 ms, which holds at camera rates, so in this model it acts as a glitch filter with no logical effect. The flip-flop captures D at the frame edge; Q is held until the next edge. Q̄ is drawn but not used or fed back.

A.4 Truth table and Boolean equation

Q = A · B · Y (gating mode). **SELECT = $Q_t + Q_{t-1} + \ldots + Q_{t-k+1}$.**

Instance	A	B	Y	Q
No motion (x = 0) or x even	0	0	any	0
x odd, x ≤ 4 (1, 3)	0	0 or 1	any	0
x odd prime ≥ 5	0 (never)	1	any	0 — no false accept
x odd composite, bin open	1 with prob. p ≥ 3/4	1	1	1 with prob. p
x odd composite, bin closed	–	1	0	0
Force mode, bin forced HIGH	–	–	1	1 for any x (guarantee lost, by design)

A.5 Error analysis

Property	Result	Basis
False accept (x ∉ L)	0 for every Y	A = 0 on primes and non-L; Q ≤ A (Thm 1)
False reject, x ∈ L, N open bins in k-frame window	$(1 - p)^N \le (1/4)^N$	Independent r per frame (Thm 2); p ≥ 3/4 (paper uses 1/2)
Default Y = 1111, k = 5 (N = 5)	≤ 0.001 (≤ 3.1% at p = 1/2)	$(1/4)^5$ and $(1/2)^5$
Time-code arithmetic	| : N never drops; & : N never rises; + : each carry loses a bin	popcount(a+b) = popcount(a) + popcount(b) − carries (Thm 3)

Caveat: the $(1 - p)^N$ bound assumes repeated trials on the same x with independent r. In live operation x changes every frame, and frames outside L give Q = 0 by design. The code's Driver-analysis panel therefore tests the independence assumption empirically instead of assuming it.

A.6 Concurrency model

The camera read runs in a worker thread (`asyncio.to_thread`). Driver A and Driver B are asyncio tasks fed by separate queues and run concurrently on each frame; the loop waits for both (`gather(qa.join(), qb.join())`), then the clock/flip-flop task runs (qc). An analyst task scores every frame against ground truth: confusion matrices, TPR/FPR/PPV/NPV, mutual information with "x ∈ L", Wilson 95% intervals, and a check that Q equals the predicted A·B·Y. Results reach the Tk GUI through a queue.Queue.

A.7 Side paths that use the circuit's clock

An **armed frame** is one whose x is even and whose previous frame's x was also even (an odd x or no-motion frame breaks the run). On each armed frame the code makes one hidden-number spot check, one higher/lower bit-search query, and one factoriser query. A verified find adds one time-code bin through *timecode_pending*, opening one extra Y bin for one frame. This is the only feedback path: it runs from the searches into Y. There is no path from Q or Q̄ back to the drivers, so the circuit's data path remains feed-forward as stated in paper §7.1.

A.8 Mapping to the paper, and required corrections

Paper statement	Status in this code	Suggested wording
§5 IR laser, reflector, dual IR detectors A/B	Not present. Detectors are camera-derived: A = motion energy, B = motion mask	State that this build is the camera implementation; the optical bench is a separate hardware variant
§5.1 conduction at 0.15 RC; logic HIGH at 1.83 RC	Implemented exactly (RC_B = 4 ms, V_BE = 0.7 V, V_IH = 3.5 V)	No change
§5.1 1/√N fluctuation ≈ 0.05% at 1 pF	Not modelled; the link is a closed-form exponential	Mark as a physical-device estimate
§6 photon Fano factor, Mandel Q, g(2)(0)	Not computed or measured in code	Mark as theory, not a measured result
§5 / §7 74HC74 at 10 Hz	Clock is the camera frame edge (≈ 30 Hz typical), not a fixed 10 Hz source	Say “clock = frame rate” for the camera build
§7.1 inverted feedback $\bar{Q} \rightarrow D$ gives 5 Hz	$\bar{Q}$ is unused; no feedback loop implemented	Present as a design analysis, not an implemented test
§1–§4 SHA-256d mining, TEST_BITS, extranonce2	Not in this file; “mined” here counts armed frames only	Cite the mining source file separately, or remove from this system description
§7 step 6: false reject $(1/2)^k$	Implemented as $(1 - p)^N$, N open Y bins, $p \geq 3/4$	Use $(1 - p)^N$ and state $p \geq 3/4$